

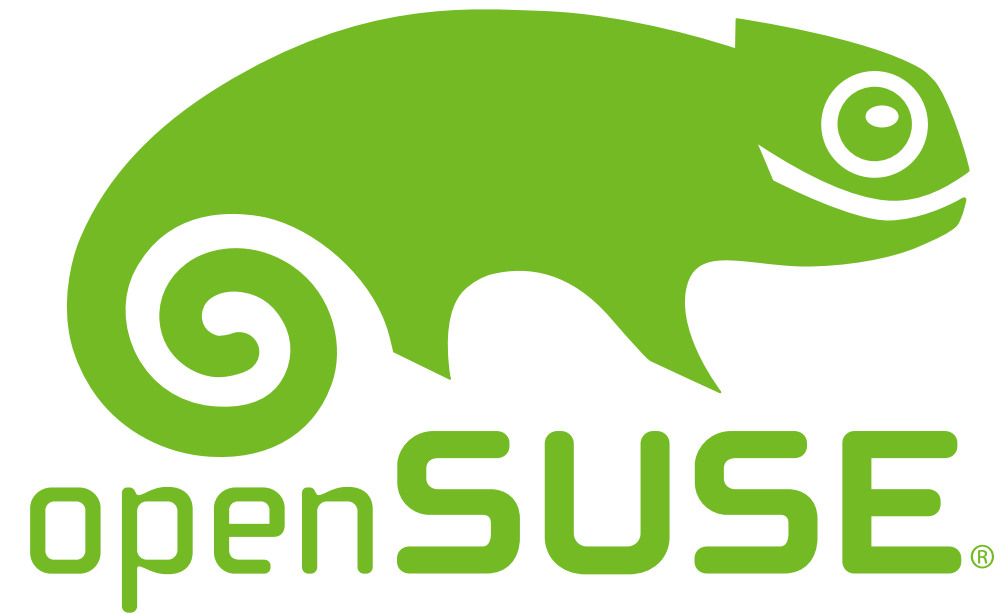
Green Computing from the CLI

Measuring Energy Consumption on Linux 🟢 🐧

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About Me

- Veteran Unix Admin / BOFH background
- QA Engineer @SUSE
- Open Source Contributor
- OpenSUSE Package Maintainer
- 🌍 Passionate about efficiency and sustainability

Why Energy Efficiency?

It's not just "being green"

- **Mobile & IoT:** Battery life is a competitive advantage.
- **Data Centers:**
Small savings $\times$ thousands of servers = Millions of € and tons of CO_2 .
- **Performance:** High energy = Heat = *Thermal Throttling*.
- **The "Cloud" Gap:** "If I can't touch the hardware, how do I measure it?"



Basics math: Power vs. Energy

Power [Watt]:

The *rate* at which energy is consumed.

Analogy: Car speed (90 km/h).

Energy [Joule]:

The *total* amount used over time.

$$\text{Energy} = \text{Power} \times \text{Time}$$

Analogy: Distance traveled (200 km).

Our Goal: Minimize the total Joules for a specific task.

The Hardware Magic: Intel RAPL

Running Average Power Limit

Modern CPUs (Intel/AMD) don't have physical "meters" inside, but they have very accurate **Power Models**.

- **Domains:**
 - **PKG:** Entire CPU socket.
 - **CORE:** Computation cores.
 - **DRAM:** Memory controller & RAM.
- **Accuracy:** Not a "truth" but a highly precise estimation based on hardware counters, voltage, and temperature.

The Linux Interface: powercap

Linux exposes RAPL data via the `powercap` framework in the virtual filesystem:

```
/sys/class/powercap/intel-rapl/
```

```
# A "quick and dirty" measurement script
def read_energy():
    with open("/sys/class/powercap/intel-rapl:0/energy_uj", "r") as f:
        return int(f.read())

v0 = read_energy()
time.sleep(1)
v1 = read_energy()
print(f"Average Power: {(v1-v0)/1e6:.1f} W")
```

The CLI Toolbox

Tool	Best For...	Command
powertop	System-wide diagnosis & "vampire" processes.	<code>sudo powertop</code>
powerstat	Real-time monitoring of system drain.	<code>sudo powerstat</code>
perf	Precision surgical measurement of a command.	<code>sudo perf stat -e power/energy-pkg/</code>
s-tui	Visualizing stress vs. power vs. frequency.	<code>s-tui</code>

Experiment 1: sort a 1GB text file

let's create a 1GB text file

```
$ tr -dc 'A-Za-z0-9 ' < /dev/urandom | fold -w 80 | head -c 1G > big.txt
$ wc -l big.txt
13256071 big.txt

$ sort big.txt > /dev/null
```

Takes ~25 seconds.

We could sort using parallelism, but Parallel sorting uses more cores (more Watts).
Will it also consume more Energy (Joules)?

```
$ time sort --parallel=16 big.txt > /dev/null
```

Takes ~6.4 seconds.

Measure 1: sort single-thread

```
$SORT_CMD1=sort --parallel=1 big.txt > /dev/null  
sudo perf stat -e power/energy-pkg/ -- $SORT_CMD1
```

Measure 2: sort multi-thread

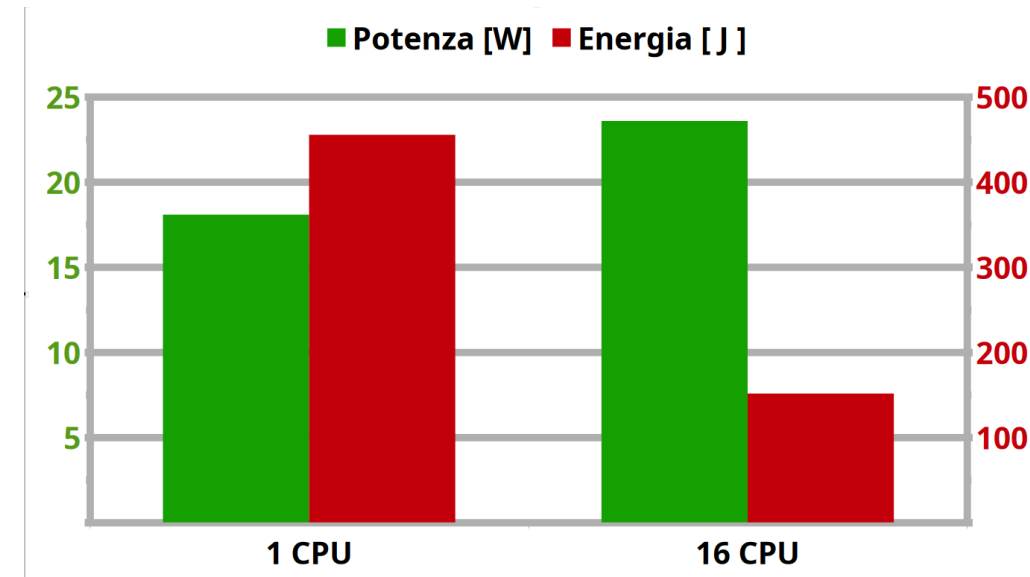
```
$SORT_CMD16=sort --parallel=16 big.txt > /dev/null  
sudo perf stat -e power/energy-pkg/ -- $SORT_CMD16
```

Experiment results

- Single-threaded: 25s @ ~18W = 455 Joules
- 16 Threads: 6.4s @ ~24W = 151 Joules

Result: ~66% Energy Saving!

Lesson: Finishing fast and letting the CPU return to "Idle" (C-states) is often the best strategy.

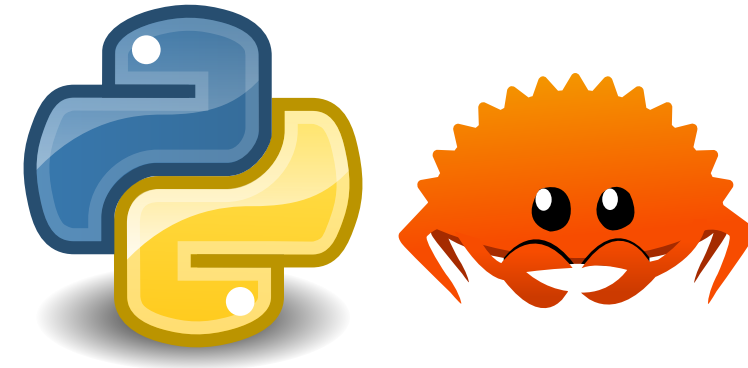


Experiment 2: Calculate the 1,000,000th prime number.

Language	Time	Energy (J)
Python 🐍	50.2 s	934 J
Rust 🦀	2.1 s	39 J

For this CPU-bound task, Rust is 23x more energy-efficient.

Note: We used the exact same algorithm for both.

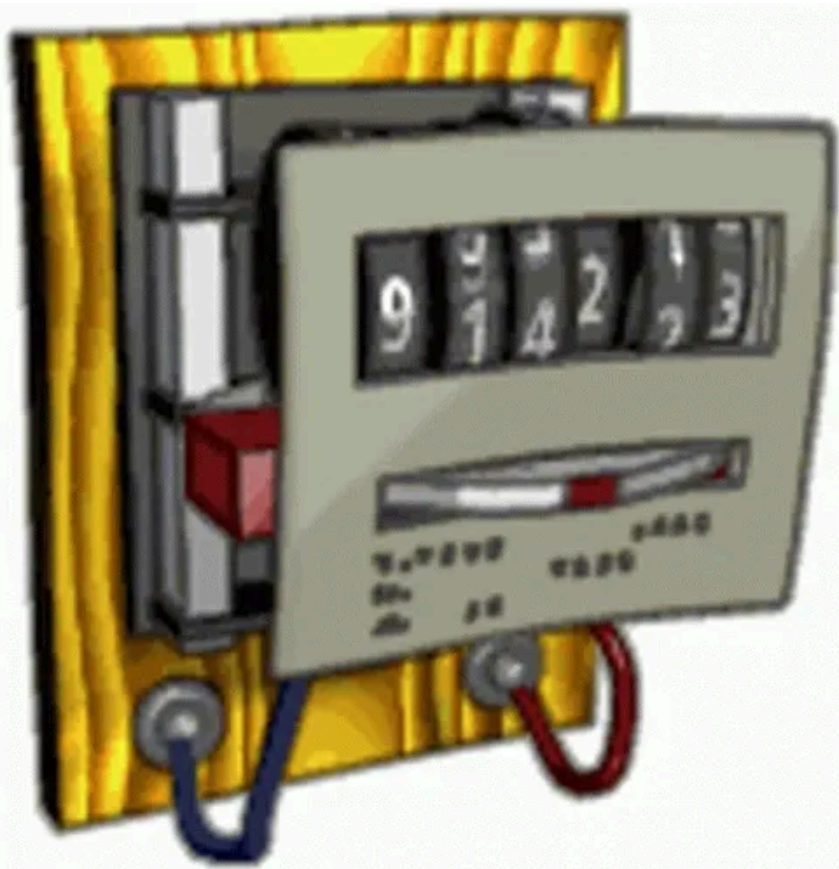


Beyond Local: The cloud native challenge

The Problem: In AWS/GCP, the hypervisor hides RAPL. `/sys/class/powercap` is empty!

a solution: Kepler (Kubernetes-based Efficient Power Level Exporter)

- Uses **eBPF** to watch CPU instructions, cache misses, and context switches.
- Uses **Machine Learning** to "guess" power consumption per Pod/Container.
- **Goal:** "Joules per Request" in your Grafana dashboard.



The Modern Stack: Scaphandre

A specialized metrology agent written in **Rust**.

- **Bridge:** Connects low-level metrics (RAPL, NVIDIA) to Prometheus.
- **Embodied Carbon:** Can estimate the energy it took to *manufacture* the server, not just run it.
- **Transparency:** Makes energy a "first-class citizen" alongside CPU and RAM metrics.

Standards & Action

OpenTelemetry (OTel)

Energy is being standardized! OTel Semantic Conventions mean your standard APM (Datadog, Jaeger) will soon show Energy by default.

Carbon Aware SDK (Green Software Foundation)

- **Spatial Shifting:** Run the job in the region with the cleanest grid (e.g., Sweden vs. Poland).
- **Temporal Shifting:** Wait 3 hours to run the batch job when the wind is blowing.

The Jevons Paradox (The Rebound Effect)

Why efficiency isn't enough to save us.

As technological progress increases the efficiency with which a resource is used (reducing the amount necessary for any one use), the rate of consumption of that resource rises due to increasing demand.

- **Example:** We moved from power-hungry spinning HDDs to highly efficient SSDs. Did we save energy? No, we just started storing 100x more data (4K video, massive databases) because it became cheap and fast.
- **The AI Era:** More efficient GPUs don't mean less energy used; they mean we train *larger* and *more complex* models.
- **The Takeaway:** Technical efficiency must be paired with conscious consumption.

Limitations & Best Practices

- **What's missing?** RAPL doesn't (usually) measure GPUs, Disks, or Network cards.
- **Repeatability:** Close your browser/Slack before measuring! Establish a baseline.
- **Model vs. Truth:** It's an estimation, but it's consistent for comparison.
- **Don't Over-optimize:** Sometimes the energy to *refactor* the code is more than the energy saved in 10 years of execution.

What Can YOU Do?

Green computing isn't just for kernel developers.

1. **Extend Hardware Lifespan:** ~70-80% of a laptop's lifetime carbon footprint is **Embodied Carbon** (manufacturing). Keep your devices for 5 years instead of 2.
2. **Embrace Dark Mode:** On modern OLED screens, black pixels are physically turned off, saving battery and energy.
3. **Temporal Shifting at Home:** Schedule heavy tasks (system updates, backups, massive downloads) overnight when grid energy is usually cleaner and cheaper.
4. **Demand Transparency:** Ask your cloud providers and software vendors about their energy metrics and sustainability goals.

Summary & Takeaways

1. **Efficiency = Quality:** Green code is often just *good* code.
2. **Measure, Don't Guess:** Use `perf stat` for surgical benchmarks.
3. **Race to Sleep:** Optimization usually beats under-clocking.
4. **Cloud is Possible:** Tools like **Kepler** are bringing transparency to shared hardware.
5. **Standardize:** Look into **OpenTelemetry** for long-term monitoring.

Thanks for your attention!

Andrea Manzini

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Questions?

- *Does it work on ARM?*
- *GPU measurements?*
- *CI/CD integration?*

